

EDA Of Airbnb(NYC) Dataset

- What Does Dataset Contain , What Patterns Had Been Extracted Helping Me For Further ML Modelling



Introduction — What is the dataset?

- Dataset Is **NYC Airbnb Open Data** Consisting Of Instances Or Operations Applied On **Airbnb** Platform.

Dataset

- NYC Airbnb Open Data

Location

- New York City

Listings

- ~49K

Features

- 16 Original Features

Period

- 2011 To 2019

Target Variable

- price

Feature Categories

Host Information

- 1) *host_id*
- 2) *host_name*
- 3) *calculated_host_listings_count*

Location

- 1) *neighbourhood_group*
- 2) *neighbourhood*
- 3) *latitude*
- 4) *longitude*

Review Information

- 1) *number_of_reviews*
- 2) *last_review*
- 3) *reviews_per_month*

Listing Information

- 1) *room_type*
- 2) *minimum_nights*
- 3) *price*
- 4) *availability_365*

Purpose

- Understand listing characteristics and prepare the data for ML modelling .
- Executing Exploratory Data Analysis On **NYC Airbnb dataset** By Using Python Libraries Helping In Understanding Data And Cleaning That Data To Be Ready For Modelling And Gaining Best Performance Can Be Expected In User Work Environment.

Feature Descriptions

id → Unique identifier for each Airbnb listing.

name → Name/title of the Airbnb listing.

host_id → Unique identifier of the host who owns/manages the listing.

host_name → Name of the host.

neighbourhood_group → The larger NYC Place(Neighbourhood) where the listing is located: Manhattan, Brooklyn, Queens, Bronx, or Staten Island.

neighbourhood → The specific neighborhood where the listing is located.

latitude → Geographic latitude of the listing.

longitude → Geographic longitude of the listing.

room_type → Type of accommodation offered: Entire home/apt, Private room, or Shared room.

minimum_nights → number of nights that a guest must book.

number_of_reviews → Total number of reviews received by the listing.

last_review → Date of the most recent review for the listing.

reviews_per_month → Average number of reviews received per month.

calculated_host_listings_count → Number of Airbnb listings associated with that host.

availability_365 → Number of days during the year that the listing is available for booking, from 0 to 365

price → Price of the listing per night in USD.

Data Quality

- **Missing values**

- name
- Host_name
- last_review
- review_per_month

- **Duplicate records**

- Zero Duplicate Instances

- **Non-Necessary Features**

- id
- host_id

- **Data types**

- 8 Numerical Features
- 6 categorical Features

EDA Analysis

◆ Data Understanding Of Price Feature

Metric

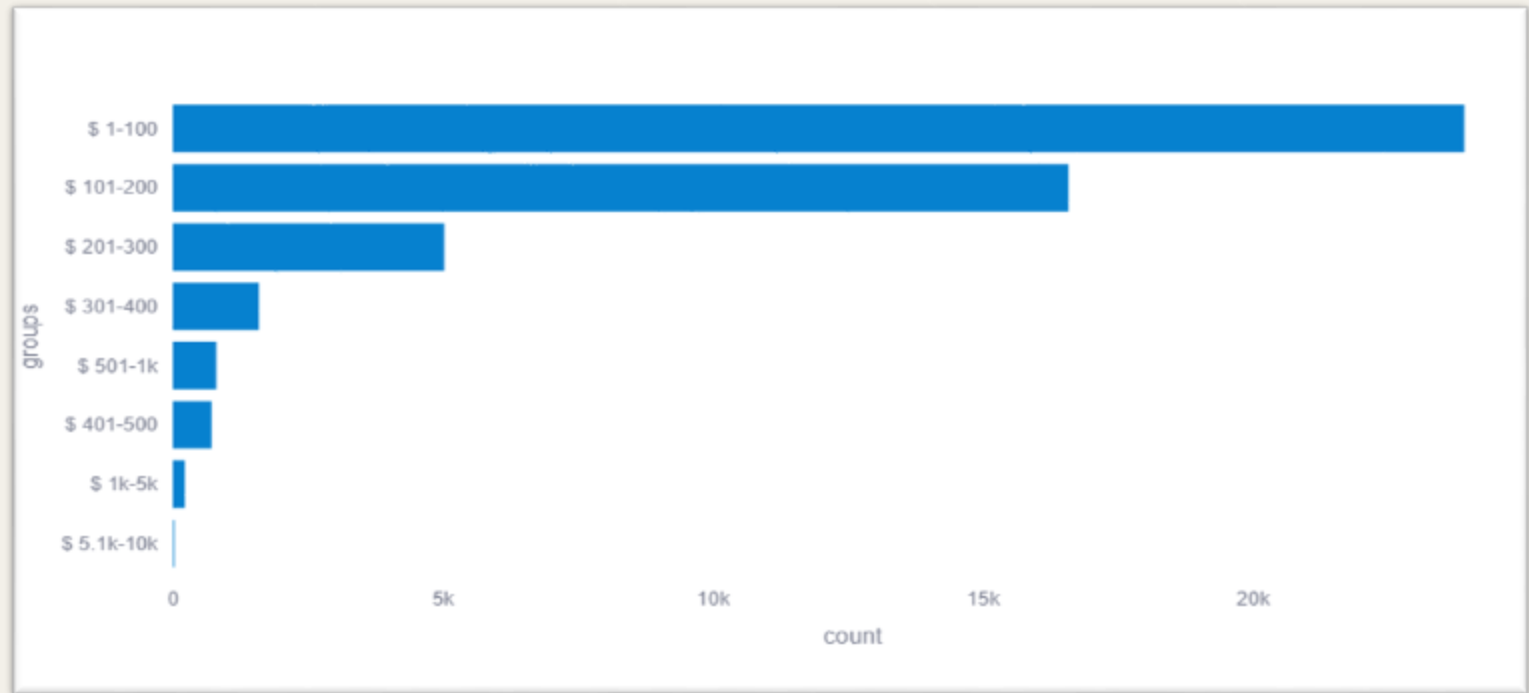
- ✓ Eleven Inconsistent Outliers Is Assigned with 0 Renting Price

Metric

- ✓ Price Distribution Is Right Skewed With Rate 19.12.

Metric

- ✓ Median Price is 106.0
- ✓ Average Price is 152.72



🔍 Key Insights

- Minimum Value In Price is **Zero Renting Prices** Which It Is Not Logic Value And Considered As Inconsistent Outliers Where Eleven Instances Contain.
- The Largest Group **1-100** With **~24k** Listings And Lowest Group Is **5.1k-10k** With **20** listings.
- The Price Is Highly Right-skewed Distributed With most Listings Concentrated At Lower Prices

EDA Analysis

◆ Data Understanding Of name Feature

Metric

- ✓ *There Is 16 Missing Values*

Metric

- ✓ *Most Frequent Description Is **Hillside Hotel***

Extracted Features

1. *desc_len*
2. *desc_lang*
3. *has_prop_keyword*
4. *has_room_keyword*
5. *has_rental_keyword*
6. *has_amenity_keyword*
7. *location_keyword*
8. *market_keyword*

What's Happened In Data Cleaning Process?

- *There Is Special Characters In Descriptions Of Listings And All Had Been Deleted To Show Those Descriptions In Clearer View.*

Key Insights

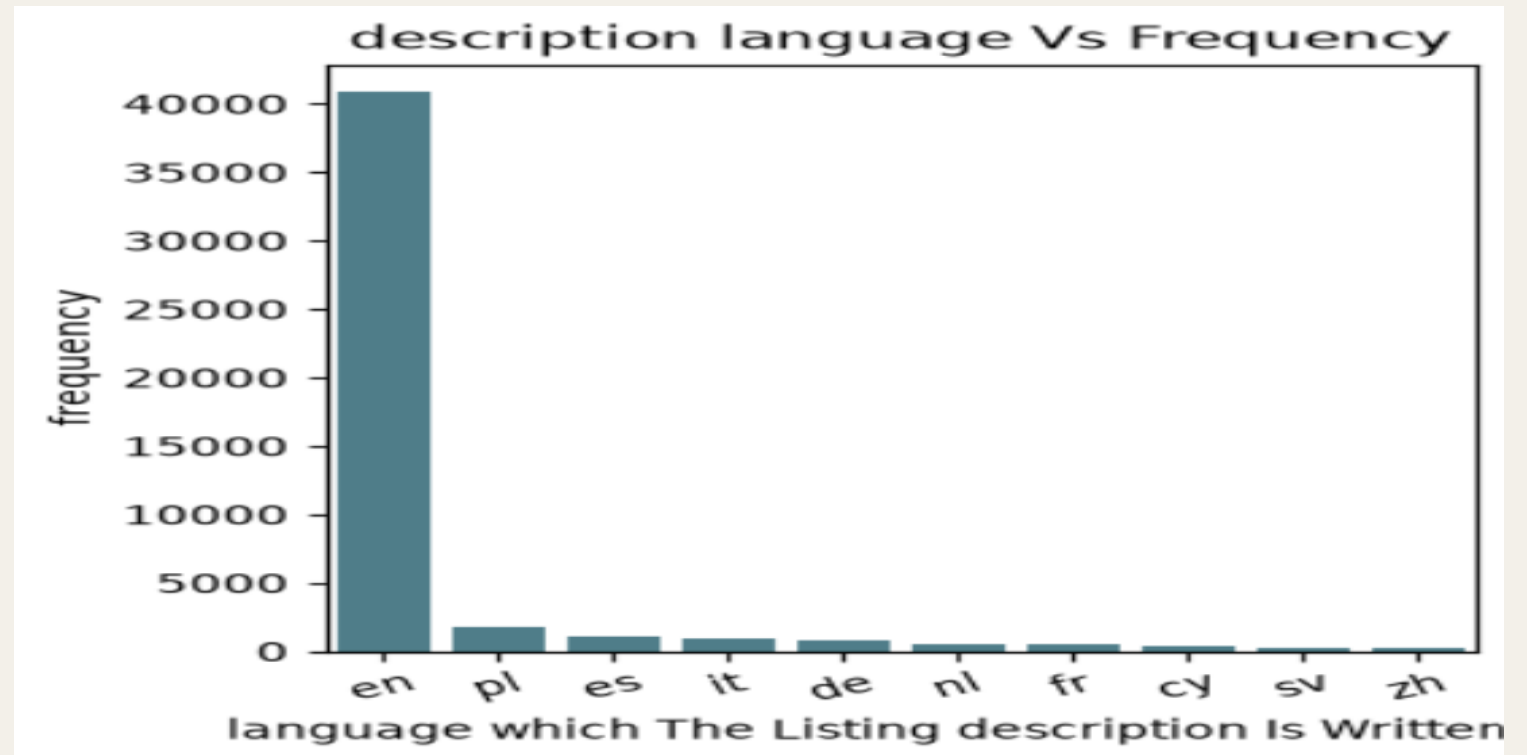
- *The Description Of Accommodations Owned Highest Renting Costs In That Dataset Are*
 - ***one-BR Lincoln Center.***
 - ***Furnished room In Astoria***
 - ***Luxury one bedroom***
 - ***Stunning Manhattan Views***

EDA Analysis

◆ Data Understanding Of desc_lang Feature

Metric

- ✓ *Description Of Listings Is Written In 71 Variant Languages*



🔍 Key Insights

- ~ **41k** Listings Description Are Written In **English Language** and ~ **2k** In **polish language**

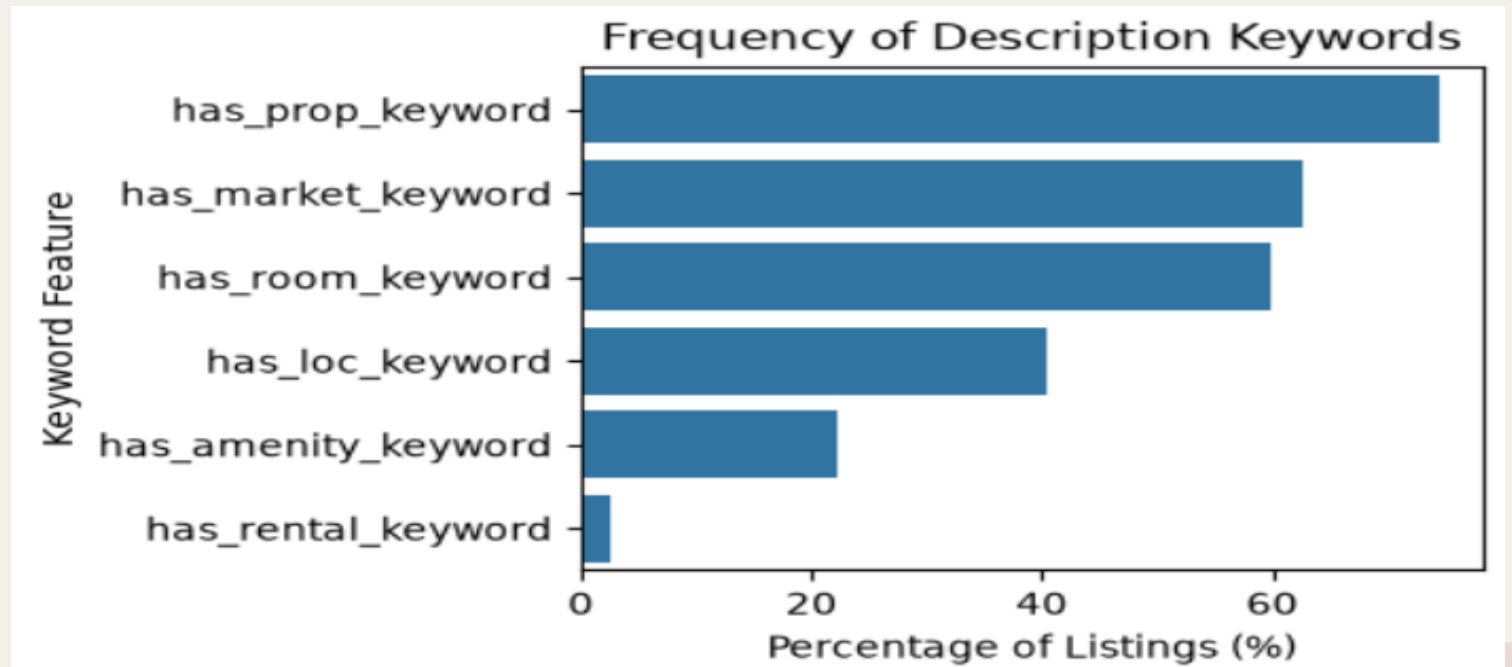
EDA Analysis

◆ Data Understanding has keywords Features

Feature	View Percentage
has_prop_keyword	74.5%
has_market_keyword	62.6%
has_room_keyword	60.0%
has_loc_keyword	40.3%
has_amenity_keyword	22.4%
has_rental_keyword	2.5%

Property Keywords

'apartment', 'apt', 'flat', 'home', 'house', 'room', 'studio', 'loft', 'suite', 'condo', 'condominium', 'townhouse', others...



🔍 Key Insights

- Property, market, and room-related terminology appears frequently in listing descriptions, while explicit rental terminology is relatively uncommon And Which It Will Help While ML Modelling

EDA Analysis

◆ Data Understanding Of host_name Feature

Metric

- ✓ *There Is 21 Missing Values*

Metric

- ✓ *Unique Hosts Is 11452 Name*

Metric

- ✓ *Most Frequent Host Is **Michael** with 417 time*

Extracted Features

1. *host_count*

What's Happened In Data Cleaning Process?

- *There Is Special Characters In Host Names And All Had Been Deleted To Show Those Names In Clearer View.*
- *Replacing All Nan Values With 'Unknown'.*

Key Insights

- *The Most Frequent Ten Hosts In That Dataset Provided Renting Of Their Houses Was Just Individual Persons. (One Host Raised Rentals On Airbnb In US New York City) Are Presented In Visualized Charts In Jupyter Python Notebook.*
- *Multiple Hosts Provided Rentals Greater Than Single Hosts Where Number Of Rentals Provided By Multiples Are **48587** Listings While Singles **296**.*



Mapping Listing Locations

Key Insights

- *The Listing Locations Are Spread In All Five Boroughs At New York City*

EDA Analysis

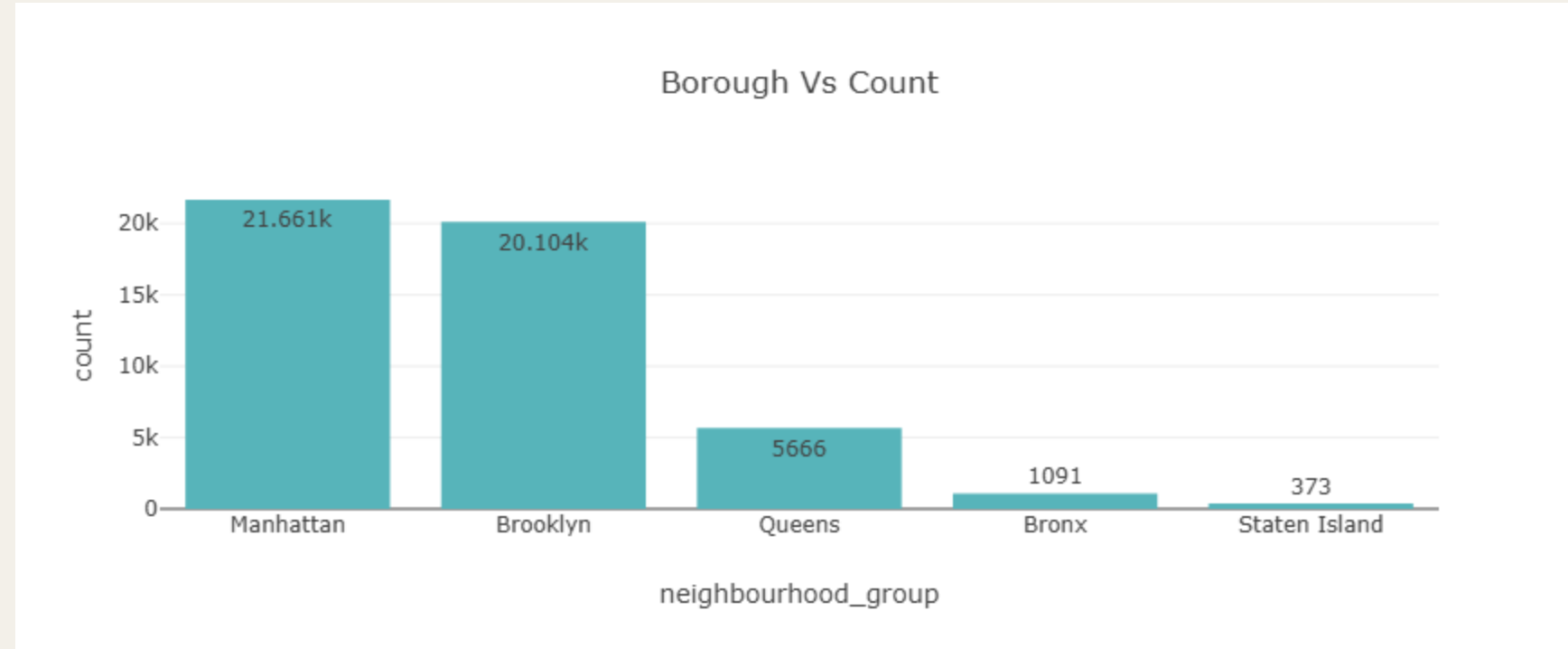
◆ Data Understanding Of neighbourhood_group Feature

Metric

✓ *There Is No Missing Values*

Metric

✓ *There Are Only Five Unique Boroughs*



🔍 Key Insights

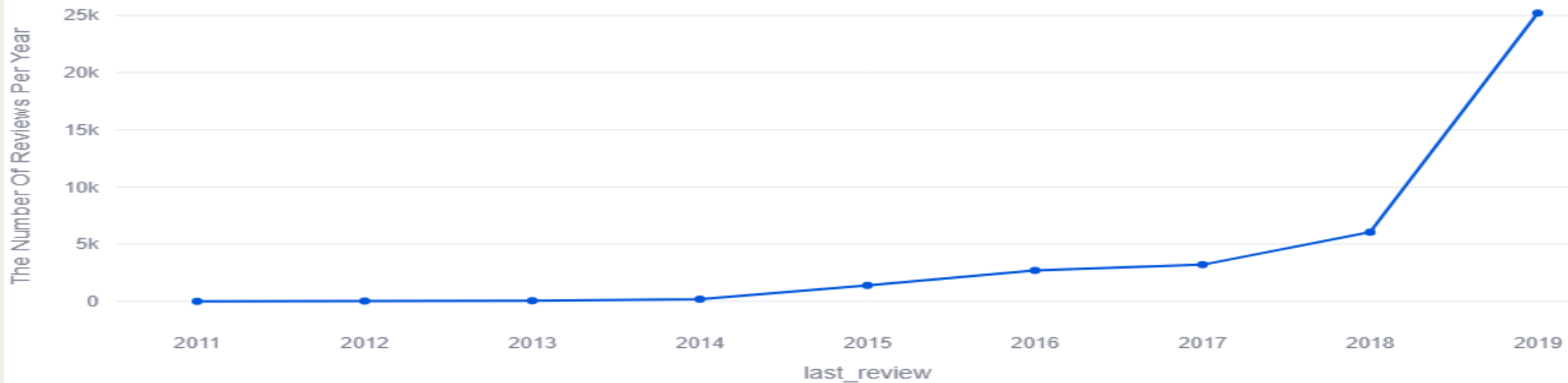
- *This Feature Is A Categorical Variable Consisting Of Five Boroughs (Manhattan , Brooklyn , Queens , Bronx And Staten Island)*
- *Most Two Boroughs Where Rentals Number Is Highest Are **Manhattan** And **Brooklyn** And Therefore Airbnb Activity Is Concerned With Two Previous Boroughs*

Listings Number Progress Over Years

Key Insights

- Number Of Listings Increases Over Years From 2011 To 2019 Where Highest Number Of Listings Was Provided In Year 2019.

Year Versus count of All reviews For Each Year



EDA Analysis

◆ Data Understanding Of room_type Feature

Metric

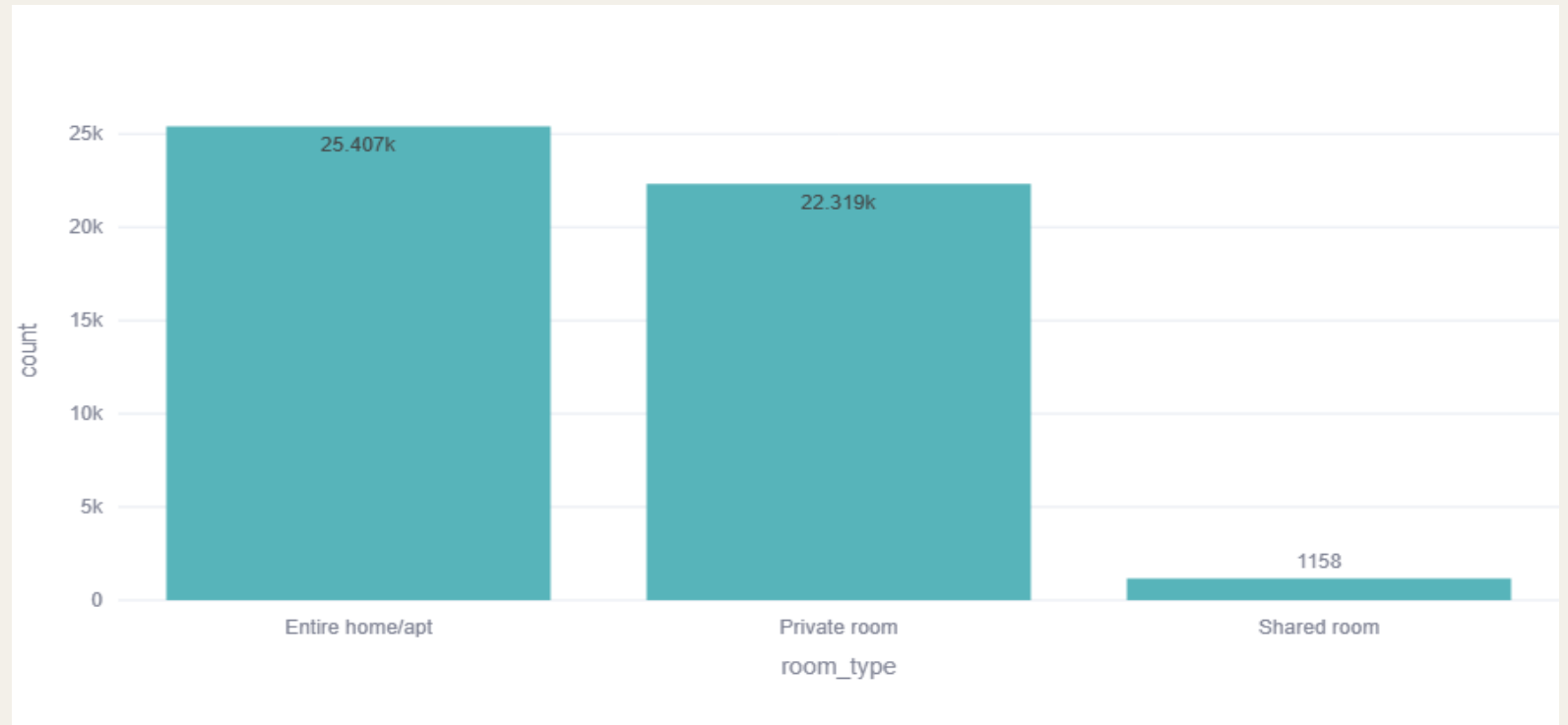
- ✓ *There Is No Missing Values*

Metric

- ✓ *There Is Three Unique Room Types*

Metric

- ✓ *Most Frequent Room Type Is Entire Home*



🔍 Key Insights

- The Most Frequent Room Type Is **Entire home/apt** Where The Average Of Renting Price Is **212.81 USA Dollar** And Maximum Price Was In That Room Type Is **10 USA Dollar**

EDA Analysis

◆ Data Understanding Of room_type Feature

Metric

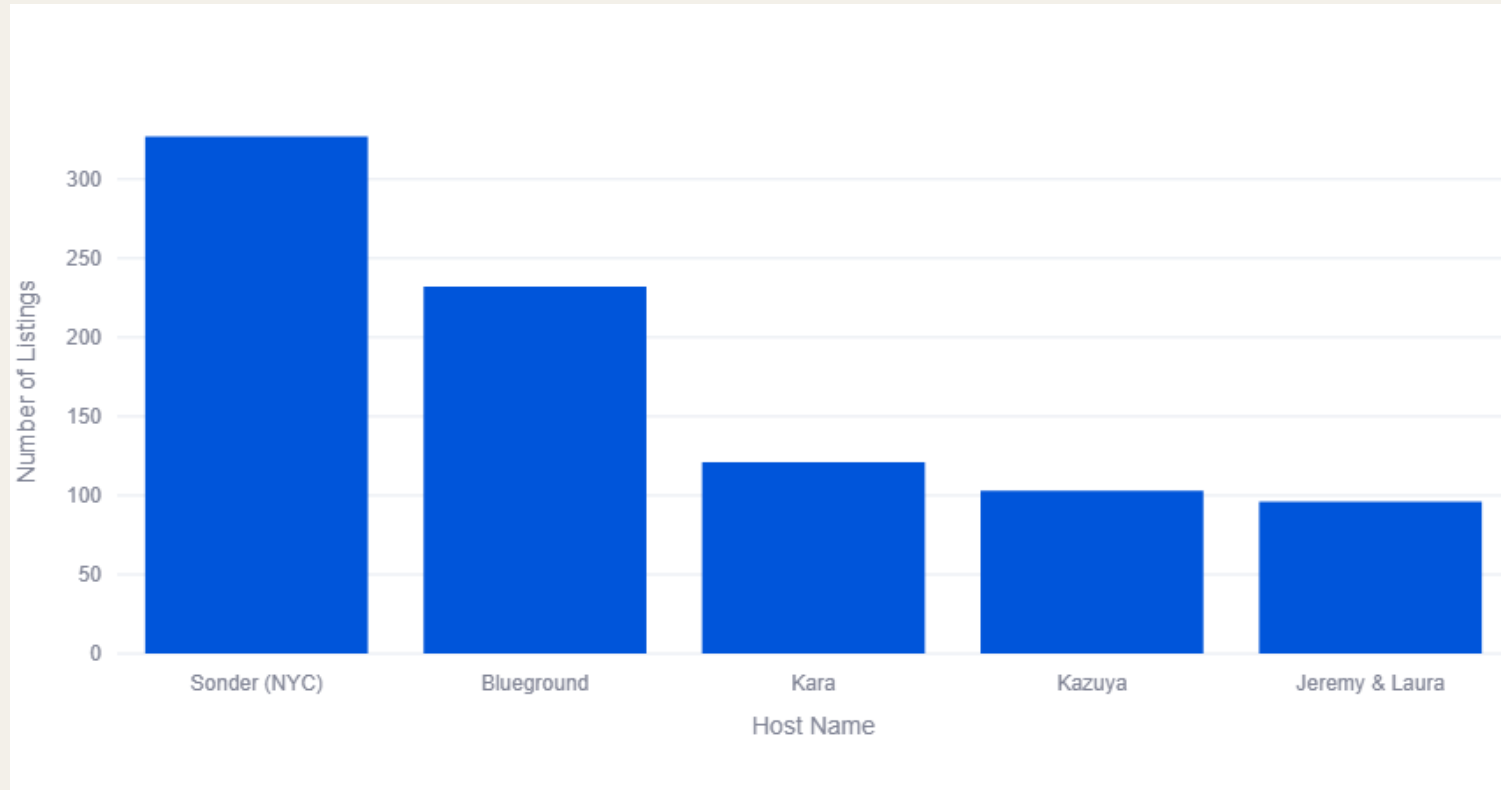
- ✓ *There Is No Missing Values*

Metric

- ✓ *Less Number of Listings Which Host Provided Is 1 accommodation*

Metric

- ✓ *Maximum Number Of Listings Which Hos Provided Is 327 Listings*



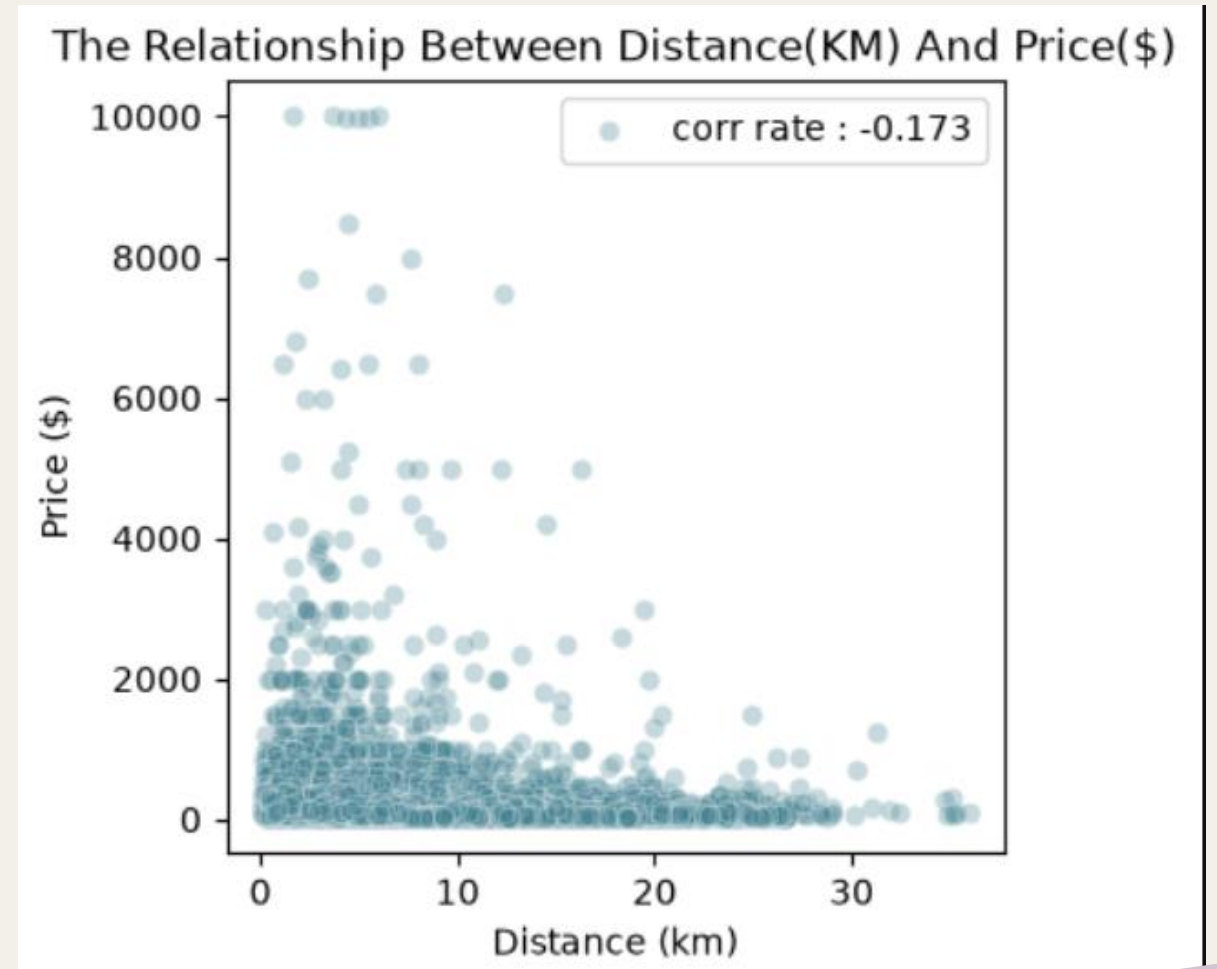
🔍 Key Insights

- *The Host Providing Highest Number Of Listings Are **Sonder(NYC)** With **327** Renting Accommodation And **Sonder(NYC)** Is One Of Most Popular Companies Providing short-term Rentals In New York City.*

Distribution Of Distance(From City Center) And Price

Key Insights

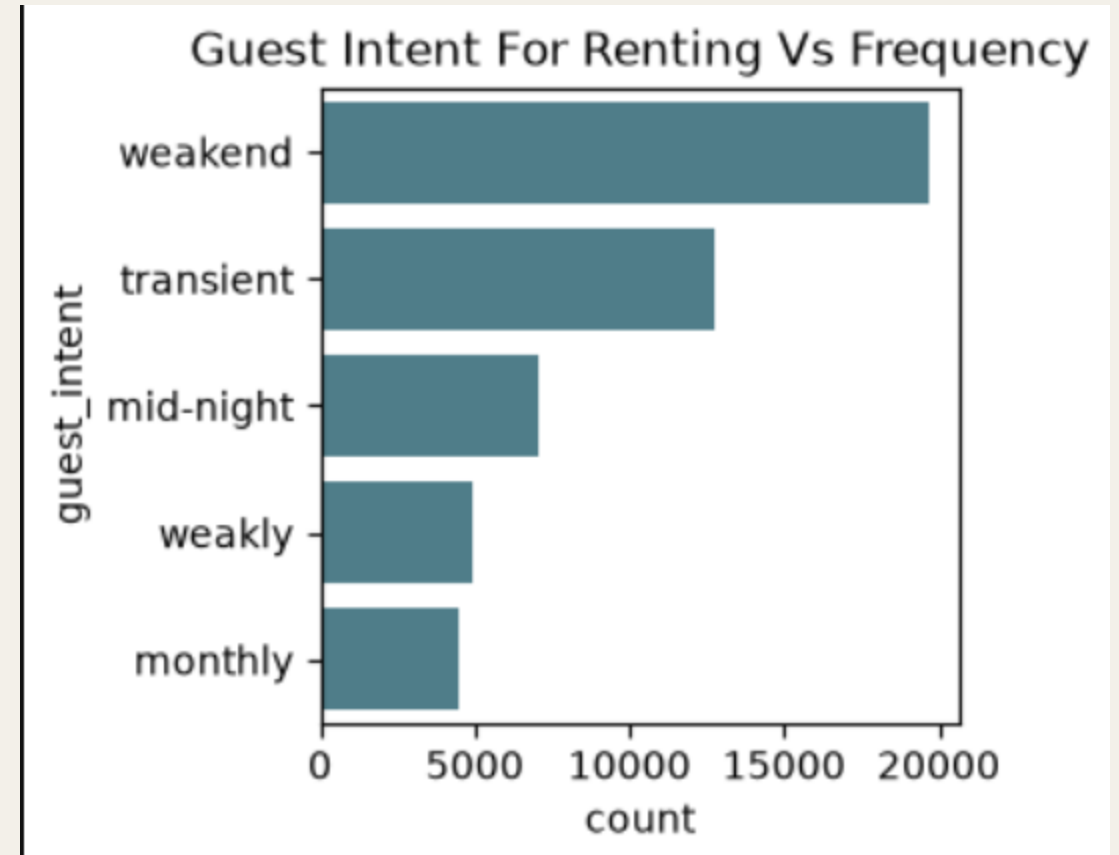
- There Is A Weak Negative Relationship Between Distance And Price With **-0.173** . Therefore , We Can't Say That The Distance_km Feature Alone Determines The Price .
- Price Tends To **Slowly Decrease** As The Distance From The City Center **Increases**



Guest Intent Frequency

Key Insights

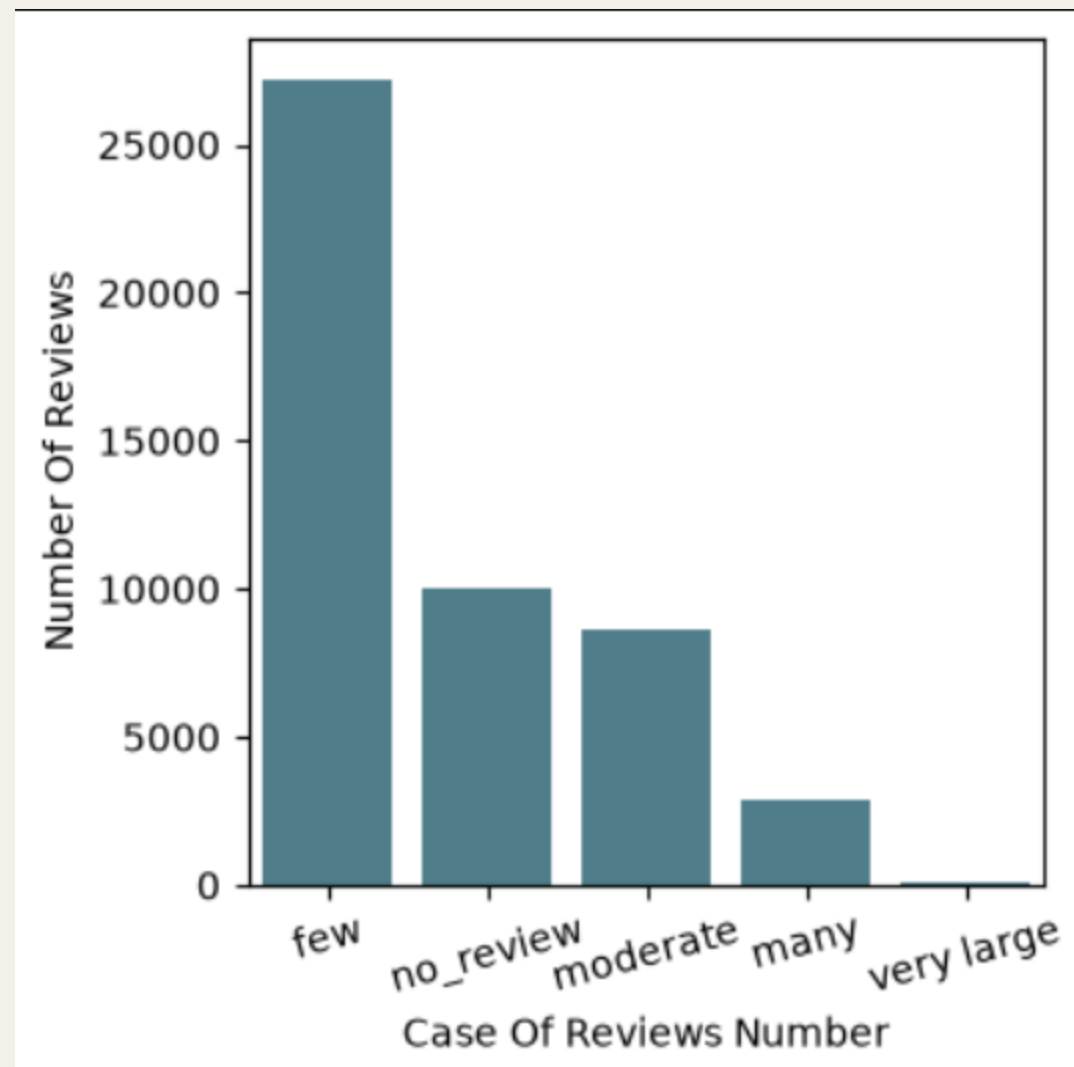
- The Dominant Intent Of Guest Is Renting For **Weekend** With **19691** Time
- **Short-term/stay-pattern categories** dominate the dataset , while **monthly-oriented listings** form a smaller group



Frequency of review Number case

Key Insights

- Huge Number of listings (accommodations) Is Categorized As **Few** In Review Category Feature At Precent **55.67%** Which It Could Represent One Of Those Probabilities :
 1. *Newer listings*
 2. *Inactive listings*
 3. *Listings with low booking activity*
 4. *Listings that simply received small number of reviews*



Insights

Set-1

EDA Insights

Explore • Visualize • Discover

- the number of instances In Renting Dataset Is **48895** While The Number Of Features After Dropping Non-Necessary Features Is **14**
- There Is No Duplicated 'Repeated' Instances In That Data .
- Features Has Missing Values Are **name , host_name , last_review And reviews_per_month** (**4 features has And 10 hasn't**) .
- There Is **8 Numeric Feature** And **6 Categorical Feature** - **Price** Feature Is **Positive Integer Numbers** Such That Minimum Value Is **Zero Renting Price** Which It Is Not Logic And Considered As Inconsistent Outliers Must Be Removed .
- There Are Eleven Inconsistent Outliers Are Valued With zeroe , And They'd been Removed From cleaned DataFrame.
- The mean Of Renting Prices Is **152.72** And The median Value Is Only **106.0** While The Maximum Value Is **10k** Is An Extreme Outlier.
- The Distribution Of Price Feature Values has **Right-Skewed Spread** With Skewness Rate **19.1** And It's not Suitable For Modelling Predictive AI Algorithm Accurately And It Must Be Normalized Before Modelling Process.

Insights

Set-2

EDA Insights

Explore • Visualize • Discover

- The Largest Group Is **1 - 100** With **23.917k Listings** And Lowest Group Is **5.1k - 10k** With **20 listings**.
- **name** Feature is A Categorical Feature Have **47905** Variant Descriptions Of Listing On Airbnb Website In New York City Had Been Done.
- The Most Repeated Characteristics Of Renting Which They Had Been Listed In That Dataset Is **Hillside Hotel**.
- The Description Of Booking Which It Owned Highest cost Is **1-BR Lincoln Center , Furnished room in Astoria And Luxury 1 bedroom app-stunning mangattan views** - **host_name** Feature is A Categorical Feature Have **11452** Variant Hosts For Renting Accomodation On Airbnb Website In New York City Had Been Shown. Is **Zero Renting Price** Which It Is Not Logic And Considered As Inconsistent Outliers Must Be Removed .
- The Most Repeated host On Airbnb In New York City Is **Michael** With **41** Listing.
- Number Of Hostings Which One Host had Provided Are **46758** Are While they Have More One Host Are **2137**
- The Most Repeated Ten Hosts In That Dataset Provided Renting Of Thier Houses And Was Just Individual Persons (one host Raised renting On Airbnb Website From New York City In Us) Presented In **Previous Charts In Section Data Analysis** where highest Hostings Was Are Provided By **micheal** With **417** Hostings And Lowest Ones By **Mike** with **194**

Insights

Set-3

EDA Insights

Explore • Visualize • Discover

- The Type Of Listings Which **Mike** Offered For Renting In That Data Are **Entire Home** With 72.7 % Out Of Number Of Listings Was Offered By Host **Mike** And Private Room With **27.3%**
- The Most Repeated Ten Hosts In That Dataset Provided Renting Of Thier Houses And Was shared Persons (more than one host Raise renting On Airbnb Website From New York City In Us) Presented In **Previous Chart In Data Analysis Section** where highest Hostings Was Are Provided By **Sonder and Nyc** With **327 Hostings** And Lowest Ones By **(pi and leo),(charlie and selenia) , (jess and ana) And (wesly and jessica)** With **8**
- The Type Of Listings Which **sonder and nyc** Offered For Renting In That Data Are **Entire Home** With **25.0 %** Out Of Number Of Listings Was Offered By Host **sonder and nyc** And Private Room With **75%**
- The Type Of Listings Which **pi and leo** Offered For Renting In That Data Are **Entire Home** With **87.5 %** Out Of Number Of Listings Was Offered By Host **pi and leo** And Private Room With **12.5%** .
- **jess and ana** Offered Only Just Shared Room For Renting In All Cases In That Data
 - The Type Of Listings Which **wesly and jessica** Offered For Renting In That Data Are **Entire Home** With **75 %** Out Of Number Of Listings Was Offered By Host **wesly and jessica** And **Private Room** With **25 %**

Insights

Set-4

EDA Insights

Explore • Visualize • Discover

- The Highest Ten Renting Averages Provided From Individual Persons (one host Raised renting On Airbnb Website From New York City In Us) Where Largest Average Provided By **olson** With **9.999 K** While Less Average By **viberlyn** With **2995**
- **Olson** Offered Only Just Entire Home For Renting In All his Cases In That Data.
- **viberlyn** Offered Only Just Entire Home For Renting In All his Cases In That Data .
- The Type Of Listings Which **pi and leo** Offered For Renting In That Data Are **Entire Home** With **87.5 %** Out Of Number Of Listings Was Offered By Host **pi and leo** And Private Room With **12.5%** .
- The Highest Ten Renting Averages Provided From shared Persons (more one host Raised renting On Airbnb Website From New York City In Us) Where Largest Average Provided By **Jay and Liy** With **6000** While Less Average By **jeff and kelly** With **949**
- **Jay and Liy** Offered Only Just Entire home For Renting In All thier Cases In That Data.
- **jeff and kelly** Offered Only Just Private room For Renting In All thier Cases In That Data.
- **neighbourhood_night** Feature is A Categorical Feature Have **Five Different Boroughs** Are Brooklyn , Manhattan ,Queens , Staten_Island and Bronx

Insights

Set-5

EDA Insights

Explore • Visualize • Discover

- most two boroughs have largest number of listings Are **Manhattan and Brooklyn** And Therefore , Airbnb Activity Is Heavily Concerned In Those Two Previous Boroughs
- There Is No Missing Values Here In This Column
- The Most Frequent Borough Where Renting Is Highest Is **Manhattan** While Lowest Number Of Booking Renting Is In **Bronx**
- The Maximum Renting Price Was In **Manhatta , Brooklyn And Queens** And All Those Boroughs Are Coastal Cities.
- The Most Three Average Prices Of Renting Was In **Manhattan , Brooklyn And Staten Island**
- The Hugest Type Of Rooms Was Renting In All Boroughs Is **Shared Rooms** While Lowest Type Of Rooms Was Renting In All Boroughs Is **Entire Home/Appartment**
- **neighbourhood** Feature is A Categorical Feature Have **221 Different Places**
- There Is No Missing Values Here In **neighbourhood** Feature , Airbnb Activity Is Heavily Concerned In Those Two Previous Boroughs
- The Highest Place In Which Renting Numbers was hugest **Williamsburg** Located In **Brooklyn Borough** With Average Price **143.8**

Insights

Set-6

EDA Insights

Explore • Visualize • Discover

- Also According To Visualization Chart , The Highest Renting Location Was In **Manhattan**
- **Latitude and Longitude** is Numerical Features Have No Missing Value
- The Samples Of This Renting Data Set Is Spread Arroud All Five Boroughs In New York City
- The Density Of Description Lengths Is Highly Ranged From **3 to 10 words** For Almost All Listings, Even Rentals With High Prices, is ranged between 3 to 10 words.
- All Single Hosts Provided Prices Ranged From **25 to 2500** And All Description Lengths Consist Of Only One Word.
- High Priced Rentals Were Provided From **Multiple Hosts**.
- The Median Distance (km) From City Center Is **6.391 km**. Therefore, We Can't Say That The Distance_km Feature Alone Determines The Price. Price Tends To Slowly Decrease As The Distance From The City Center Increases.
- A Huge Number Of Listings (Accommodations) Is Categorized As Few In The Review Category, At A Percentage Of **55.67%**. This Could Represent One Of The Following Possibilities: (Newer Listings, Inactive Listings, Listings With Low Booking Activity, Or Listings That Simply Received A Small Number Of Reviews).

Utilities Of EDA ?

Pandas

Numpy

Matplotlib

Seaborn

Plotly

datasist

num2words

Dependencies Of Deployment ?

os

io

base64

PIL

Streamlit

Github And Deployment Link

- Deployment Link :
 - [EDA-Airbnb NYC Dataset · Streamlit](#)
- Github Link :
 - [EDA airbnb-NYC/presentation-links at main · Islam-Yousri/EDA airbnb-NYC](#)

